

INTEGRATED IT FUNDAMENTALS · IMPLEMENTATION HANDBOOK

Cloud, Database & Cybersecurity Essentials

A step-by-step guide to running every lab, checklist, and capstone activity in this Azure training kit — framed for a Nigerian petroleum-regulator audience.

Companion to the GitHub repository

Use alongside the README and the .md lab, dataset, and capstone files.

Microsoft Certified Trainer edition

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SECTION 01

Introduction & How to Use This Handbook

This handbook turns the MCT Delivery Kit — "Cloud, Database & Cybersecurity Essentials: Integrated IT Fundamentals Training" — into a step-by-step implementation guide. The kit is a complete 5-day (40-hour) instructor-led course, delivered entirely on Microsoft Azure, for IT staff at a Nigerian petroleum/oil & gas regulatory agency (think NNPC Ltd, NUPRC, or NMDPRA). The audience is a mix of system administrators, DBAs, DevOps engineers, and technical managers — intermediate IT level, mixed cloud maturity, SQL helpful but not mandatory.

This handbook is written for two readers at once. If you are the MCT (Microsoft Certified Trainer) delivering the course, follow the setup steps before Day 1 and use the day-by-day sections to run each lab live. If you are a participant, follow the same numbered steps at your own workstation during the hands-on blocks — everything here matches the actual lab guides and datasets in the repository, so nothing you read here should surprise you when you open the files yourself.

Every example in this handbook uses one running scenario: a Nigerian downstream/midstream petroleum regulator tracking licensing, crude oil production reporting, downstream fuel distribution, field inspections, environmental compliance, and revenue/royalty data across depots and terminals in Lagos, Port Harcourt, Warri, and Kaduna. Keeping one story running for all 5 days is deliberate — it is exactly how the course kit is built, and it is why the Day 5 capstone feels like the natural conclusion of the week rather than a bolt-on exam.

The 5-day arc at a glance

- Day 1 — Cloud Fundamentals Recap: IaaS/PaaS/SaaS, deployment patterns, and a cloud service selection case study (no resources deployed).
- Day 2 — Database Fundamentals: create a free-tier Azure SQL Database, deploy a normalized petroleum inventory schema, practice CRUD SQL.
- Day 3 — Database Administration & Performance: diagnose a slow query with execution plans, fix it with an index, then review backup/PITR and geo-replication.
- Day 4 — Cybersecurity Essentials: harden the Day 2 database — identity (Entra ID/RBAC/MFA), encryption, network controls, Defender for SQL, auditing — and complete a 24-control security checklist.
- Day 5 — Integrated Lab & Capstone: build a small end-to-end app + managed database + security controls, submit 3 deliverables, and defend the build in a short oral exam.

Who should read which parts

- MCT instructors: read Sections 1–3 fully before the course starts, then read the matching day's section the night before delivery, plus Section 9 (Instructor Delivery Guide).
- Participants: read Section 1–3 once before Day 1, then follow the Day-N section as you work through each day's hands-on lab.

- Everyone: keep Section 10 (Troubleshooting & FAQ) and Section 11 (Cost management & Glossary) open throughout the week.

TIP This handbook assumes you already have the course repository cloned to your laptop. If you do not, clone or download it now — every step below refers to real files inside the repository (labs, datasets, capstone templates) by their actual filenames.

Prerequisites for the week: an Azure account able to create resources (individual free account strongly preferred — see Section 3), Azure Data Studio or SQL Server Management Studio installed, basic familiarity with IT concepts, and a willingness to think in terms of the regulator's real duties — data sovereignty, NDPR/NDPA 2023 compliance, audit trails, least privilege, and keeping critical systems available.

IMPORTANT This is a training kit, not a production deployment guide. Every lab uses free-tier or basic-tier Azure resources, synthetic data only, and a single shared, disposable resource group per learner. Do not put real licensee, production, or royalty data into any lab resource — see Section 3's cost-safety guardrails and Section 11's cleanup checklist.

SECTION 02

Repository Structure & File Map

The repository is organized so each day of the course has exactly one lab guide and one (or more) dataset file, plus a facilitation plan that ties everything together and a capstone folder for Day 5. Before you touch any lab, skim this map once so you know where to find the file you need without hunting.

File map

Path	What it is	When you use it
README_START_HERE.md	Top-level orientation: what's in the kit, Azure environment setup, cost guardrails, contingency plan.	Read first, before anything else.
01_Facilitation_Plan.md	The instructor's minute-by-minute playbook for all 5 days: schedules, talking points, misconceptions, prep checklists, contingency, confidence primer.	Instructors: read end-to-end 2–3 weeks out, then the day's section the night before.
slides/Cloud_DB_Security_Essentials_Facilitator_Deck.pptx	The 20-slide presentation deck for lectures.	During each day's lecture blocks.
labs/Lab_Day1_Cloud_Service_Selection.md	Day 1 lab: cloud service-selection case study, decision matrix, instructor model answer.	Day 1 afternoon hands-on lab.
labs/Lab_Day2_Database_Fundamentals_CRUD.md	Day 2 lab: create the free Azure SQL Database, deploy the inventory schema, run CRUD SQL.	Day 2, after the morning teaching blocks.

Path	What it is	When you use it
labs/Lab_Day3_DBA_Performance.md	Day 3 lab: execution-plan analysis, index fix, backup/PITR + geo-replication walkthrough.	Day 3 afternoon hands-on lab.
labs/Lab_Day4_Security_Hardening.md	Day 4 lab: identity, encryption, network hardening, Defender for SQL, auditing, checklist.	Day 4 afternoon hands-on lab.
labs/datasets/petroleum_inventory_schema.sql	T-SQL: 5-table normalized inventory schema + 74 seed rows across 4 cities.	Day 2 Part 3 (deploy) and referenced again Day 4.
labs/datasets/petroleum_inventory_seed.csv	CSV version of movement/stock data, for import-exercise variants.	Optional Day 2 alternative import practice.
labs/datasets/day3_performance_lab.sql	T-SQL: builds a ~12,000-row InspectionEvents table with a deliberately slow query and its index fix.	Day 3 lab, all parts.
labs/datasets/day4_security_checklist.md	Reusable 24-control Azure security hardening checklist.	Day 4 lab Part 7, and reused unmodified as a Day 5 capstone deliverable template.
capstone/Capstone_Project_Brief.md	Day 5 brief: scenario choice, functional scope, 3 deliverables, service mapping, constraints, timeline, submission steps.	Day 5 morning kickoff and throughout the day.
capstone/Capstone_Deliverable_Templates.md	Fill-in templates for all 3 deliverables: architecture diagram, security checklist, backup/DR plan.	Day 5, while building the deliverables.
capstone/Capstone_Grading_Rubric.md	100-point weighted rubric across 4 sections + instructor scoring sheet + 70/100 pass threshold.	Day 5 grading, and to preview expectations on Day 4's wrap-up.
capstone/Oral_Defense_Guide.md	How to run the 10-minute oral defense: format, 15-question bank across 4 categories, scoring guidance.	Day 5, 16:00–17:15 defense slot.

TIP Keep the dataset files (`labs/datasets/\*.sql`, `\*.md`) copied locally before the session starts — the Facilitation Plan explicitly calls out Lagos/Abuja connectivity risk, so do not rely on re-downloading them on the day.

SECTION 03

Prerequisites & One-Time Setup

Complete everything in this section before Day 1 — ideally 5–7 business days ahead, with a final check the night before. Most first-day friction in this course comes from setup steps skipped here, not from the course content itself.

1. Get an Azure account

1. Decide whether learners will use individual Azure free accounts or a shared training subscription. Prefer individual free accounts — free-offer quotas are per-subscription, so a shared subscription exhausts fast once 10 or more learners are provisioning databases.
2. Sign up before Day 1 using "Start free" in the Azure SQL hub at aka.ms/azuresqlhub, or the general Azure free account signup.
3. Confirm the exact free-offer terms you are relying on: Azure SQL Database free offer gives up to 10 General Purpose serverless databases per subscription, with 100,000 vCore-seconds compute + 32 GB data + 32 GB backup free per month, for the life of the subscription.
4. If you prefer PostgreSQL as your primary track, confirm instead: Azure Database for PostgreSQL Flexible Server is free for 12 months on an Azure free account — 750 hours of Burstable B1MS compute + 32 GB storage + 32 GB backup per month.
5. Verify you have a valid phone number/email for identity verification, and — if using a corporate/regulator tenant — that a Global Admin has pre-approved self-service Azure free-tier signups (regulator IT security policies sometimes block this).

Source: Azure SQL Database free offer —

<https://learn.microsoft.com/en-us/azure/azure-sql/database/free-offer>

Source: Azure Database for PostgreSQL Flexible Server, free 12-month account deployment — <https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/how-to-deploy-on-azure-free-account>

2. Install the tools you need

1. Install Azure Data Studio — the recommended, cross-platform client (works well for Linux/Mac users on a mixed sysadmin team).
2. Install SQL Server Management Studio (SSMS) as a Windows-only alternative if you already know it well.
3. Confirm you have a modern browser (Edge or Chrome) for the Azure portal, and that no corporate proxy blocks portal.azure.com, *.database.windows.net, or *.postgres.database.azure.com.
4. Optional: install the Azure CLI and/or confirm Cloud Shell access if you are comfortable with the command line — not required for any lab in this kit.
5. Confirm your laptop's firewall allows outbound TCP 1433 (Azure SQL) and TCP 5432 (PostgreSQL Flexible Server). Corporate/regulator networks often block these ports by default — this is the single most common Day 2 blocker.

IMPORTANT If you are on a regulator or agency network, ask IT/security in advance whether your laptop sits behind a restrictive firewall or VPN. Request a temporary outbound allow-list for *.database.windows.net:1433, *.postgres.database.azure.com:5432, and portal.azure.com, or arrange a separate, less-restricted guest network for the training room. Test actual database connectivity from the room beforehand — a browser page loading successfully does not prove port 1433/5432 is open.

3. Clone the repository

1. Clone or download this repository to your laptop.
2. Confirm you can open labs/Lab_Day1_Cloud_Service_Selection.md and see the case study text render correctly.
3. Copy the labs/datasets/ folder to local disk (not just a cloud-drive link) — you will need petroleum_inventory_schema.sql on Day 2, day3_performance_lab.sql on Day 3, and day4_security_checklist.md on Day 4.
4. If you are an instructor with a hybrid/remote cohort, share the resource-group naming convention and dataset file links with remote learners 24 hours ahead so they are not blocked mid-session waiting for a file transfer.

4. Set the resource-group naming and tagging convention

1. Each learner uses one shared resource group for the entire course, named rg-<initials>-course (example: rg-cae-course for Chidi A. Eze).
2. Tag every resource you create with course = cdcs (short for Cloud, Database & Cybersecurity) so resources are easy to find and clean up later.
3. If you are the instructor, print or share a roster mapping learner name → initials → resource group name → subscription type. Keep it visible for quick lookups when troubleshooting.

Cost-safety guardrails — enforce these from Day 1 onward

Use the free-offer Azure SQL Database (General Purpose, serverless) for every lab database.

When the free monthly limit is reached, choose "Auto-pause the database until next month" rather than upgrading to a paid tier.

Stop or deallocate any VMs and App Service plans at the end of each day.

Keep exactly one resource group per learner: rg-<initials>-course, tagged course=cdcs.

Disconnect Azure Data Studio/SSMS Object Explorer whenever you step away, so the serverless database can auto-pause.

Do not upsize any database beyond the free-offer serverless General Purpose tier during this course.

Delete the resource group entirely at the end of Day 5 (see Section 11).

TIP Lagos/Abuja power and fibre can be unstable. Instructors: pre-build a Docker Compose bundle with mcr.microsoft.com/mssql/server and postgres images on a USB drive as an offline fallback, and test it boots on a representative learner laptop (8 GB RAM minimum recommended) before Day 1. This is insurance, not a last-minute improvisation.

SECTION 04

Implementing Day 1 — Cloud Service Selection

Day 1 has no hands-on Azure deployment — it is a structured decision exercise. The goal is to build the decision-making muscle learners will use for real on Day 5, when they actually choose and deploy Azure services for the capstone. The running example is a Crude Oil Production Reporting Portal: licensed upstream operators submit monthly production and royalty data, and field officers in Lagos, Port Harcourt, Warri and Kaduna log inspection visits against the same data.

What you'll need

1. The case study handout from labs/Lab_Day1_Cloud_Service_Selection.md, Part A.
2. A copy of the blank decision matrix (Part B of the same file) — print it or share it as a doc.
3. Optional, read-only: an Azure account signed in to browse the Azure regions page and Azure products-by-region page for reference.

Step-by-step: running the case-study lab

1. Hand out or share Part A of the lab file: the Crude Oil Production Reporting Portal requirements brief (9 numbered business requirements covering users, availability, data sensitivity, residency, team skills, budget, growth, integration, and audit).
2. Read the brief individually or in pairs (about 45 minutes). Pair a manager (business framing strength) with a sysadmin or DBA (technical depth) — this mirrors how real recommendations get made and keeps managers engaged during technical stretches.
3. Work through Matrix 1 (Compute Model: IaaS vs. PaaS), Matrix 2 (Database Service), and Matrix 3 (Azure Region) — score each option 1 (poor fit) to 5 (excellent fit) against each weighted criterion, multiply score by weight, and total each column.
4. Write one paragraph per matrix (3 total) justifying the top-scoring choice, explicitly citing the numbered requirement(s) it satisfies.
5. Write one final one-line recommendation in the form: "For the Crude Oil Production Reporting Portal, we recommend [PaaS/IaaS] compute, [database service], in [region], because ____."
6. Optional Part C (read-only Azure portal reconnaissance, no deployment): sign in to portal.azure.com, open the App Services overview blade without clicking Create, view the Pricing plan step of a Web App creation wizard and cancel out, open the Azure SQL hub free-offer banner, and browse the region picker — then close without creating anything.
7. Debrief as a group: pairs present their recommendation, and the instructor challenges assumptions with the prompt "what would change your recommendation if data residency were non-negotiable?"

Decision Matrix 1 — Compute Model (IaaS vs. PaaS), compact version

Criterion	Weight	What to consider
Matches small team's ops capacity	25%	The regulator's IT team is 4 people, strong on SQL/Windows admin, no 24/7 ops

Criterion	Weight	What to consider
Cost predictability / minimal maintenance	20%	Constrained public-sector budget favors predictable OpEx over raw performance
Ability to handle spiky monthly load	20%	Usage spikes in the first 10 business days of each month
Time-to-deploy for course/lab timeline	15%	PaaS provisions and configures faster than VM patch/build cycles
Room to scale for downstream-monitoring growth	10%	Expected to roughly double data volume within 18 months
Security patching burden	10%	Ties directly into the Day 4 patching/vulnerability theme

Decision Matrix 2 — Database Service, compact version

Criterion	Weight	What to consider
Team's existing SQL Server skill fit	25%	Existing SQL Server skills transfer directly to Azure SQL Database
Built-in backup/PITR/geo-replication	20%	Directly satisfies the audit-trail and business-continuity requirement
Cost predictability at small scale	20%	Serverless auto-pause suits a constrained, predictable budget
Patching/maintenance burden	15%	Managed PaaS database removes OS/engine patching from the team
Free-tier availability for training/lab reuse	10%	Lets the team train on the exact service they will run in production
Growth headroom	10%	Must support roughly doubled data volume within 18 months

Decision Matrix 3 — Azure Region, compact version

Criterion	Weight	What to consider
Data residency / sovereignty fit	35%	There is deliberately no Azure region physically located in Nigeria today
Latency to Nigerian users	25%	Lagos, Port Harcourt, Warri, Kaduna are the primary user locations
Service availability in the region	20%	Confirm Azure SQL, App Service, and Key Vault are all present in the region
Disaster-recovery pairing option available	10%	Needed for the Day 3/Day 5 geo-replication discussion

Criterion	Weight	What to consider
Cost tier for the region	10%	Region choice affects list pricing, not just latency

IMPORTANT The Azure region matrix is intentionally a trap. There is no Azure region physically located in Nigeria. The correct answer is not to silently pick the cheapest or closest region — it is to recognize the gap and build a documented compliance justification: contractual safeguards, Microsoft's data-protection terms, and an explicit NDPA/NDPR cross-border transfer rationale. Learners who skip this reasoning should be redirected back to requirements 3 and 4 in the brief.

TIP Instructor model answer for reference (do not distribute before the exercise): PaaS (Azure App Service) for compute, Azure SQL Database for the database, and a widely available region with strong contractual data-protection commitments (for example, an EU region) explicitly justified rather than assumed. Full reasoning is in labs/Lab_Day1_Cloud_Service_Selection.md under "Instructor Model Answer."

Expected result / checkpoints

- All 9 numbered requirements are individually addressed at least once across the three matrices.
- All three decision matrices are completed with scores, weights, and totals calculated.
- A top choice is identified in each matrix with a numeric total.
- Three justification paragraphs are written, each citing specific requirement numbers.
- If Part C was completed: no Azure resources were actually created — confirm via the Resource groups blade.

SECTION 05

Implementing Day 2 — Create the Azure SQL Database, Deploy the Schema, and Run CRUD

Day 2 is where hands-on Azure work starts. Learners provision a free-tier Azure SQL Database, connect to it, deploy a real 5-table petroleum inventory schema, and practice Create/Read/Update/Delete SQL against realistic oil & gas asset data — pipeline valves, pumps, meters, PPE, and spares across facilities in Lagos, Port Harcourt, Warri, and Kaduna. This database, PetroleumInventoryDB, is reused on Day 3 (performance) and Day 4 (security hardening), so build it carefully.

What you'll need

1. Completion of the Day 1 lab (cloud service selection concepts).
2. An Azure account with permission to create resources.
3. Azure Data Studio or SSMS installed.
4. The file labs/datasets/petroleum_inventory_schema.sql from this kit, saved locally.

Part 1 — Create the Azure SQL Database (free offer)

1. In a browser, go to aka.ms/azuresqlhub (the Azure SQL hub landing page).
2. Sign in with your Azure account, then select "Start free."
3. On the Basics tab: set Subscription to your course subscription; set Resource group to Create new → rg-<initials>-course; set Database name to PetroleumInventoryDB.
4. Under Server, select Create new: choose a globally unique Server name such as sql-<initials>-course (lowercase, no spaces), pick the region your instructor specifies, and set Authentication method to "Use both SQL and Microsoft Entra authentication." Set a SQL admin login and a strong password — record it securely, you will need it again on Day 4.
5. Leave "Want to use SQL elastic pool?" set to No, and set Workload environment to Development.
6. Under Compute + storage, select Configure database, choose General Purpose – Serverless, and confirm the free-offer banner is shown (10 free databases per subscription).
7. Set Backup storage redundancy to Locally-redundant (sufficient for lab use).
8. Go to the Networking tab: set Connectivity method to Public endpoint; leave "Allow Azure services and resources to access this server" set to No; set "Add current client IP address" to Yes.
9. Go to Tags, add course = cdc and an owner tag with your initials.
10. Select Review + create, confirm the free-offer usage is shown, then select Create. Wait 2–5 minutes for deployment.
11. Once deployed, select Go to resource and confirm the Free offer badge/usage panel is visible on the Overview blade.

Source: Azure SQL Database free offer —

<https://learn.microsoft.com/en-us/azure/azure-sql/database/free-offer>

Part 2 — Connect with Azure Data Studio

1. Open Azure Data Studio (or SSMS) and select New Connection.
2. Enter the fully qualified server name from the portal Overview blade, e.g. sql-<initials>-course.database.windows.net.
3. Set Authentication type to SQL Login and enter the SQL admin username/password from Part 1.
4. Set Database to PetroleumInventoryDB and select Connect.
5. If prompted that your client IP is not allowed, go back to the Azure portal → your SQL server → Networking → Add client IP → Save, then retry.
6. Once connected, open a new query window against PetroleumInventoryDB.

Part 3 — Deploy the schema

1. Open labs/datasets/petroleum_inventory_schema.sql in Azure Data Studio (File > Open File).
2. Read the header comments first — note the normalization design: Facilities, Suppliers, Equipment, StockItems, and StockMovements (the transactional fact table).
3. Select Run (or press F5) to execute the entire script. It creates five tables, two indexes, and seeds about 74 rows of oil & gas inventory data.
4. Run the sanity-check queries at the bottom of the script to confirm the load.

Sanity-check counts after deploying the schema

```

SELECT COUNT(*) AS FacilityCount FROM dbo.Facilities; -- expect 10
SELECT COUNT(*) AS SupplierCount FROM dbo.Suppliers; -- expect 8
SELECT COUNT(*) AS EquipmentCount FROM dbo.Equipment; -- expect 14
SELECT COUNT(*) AS StockItemCount FROM dbo.StockItems; -- expect 12
SELECT COUNT(*) AS MovementCount FROM dbo.StockMovements; -- expect 30

```

Part 4 — CRUD exercises

1. 4.1 Create: insert a new depot facility and record a stock receipt against it (see code sample below).
2. 4.2 Read: run the three sample SELECT queries — HIGH-category PPE with preferred supplier, stock-on-hand per item at a named facility, and equipment currently Faulty or UnderMaintenance.
3. 4.3 Update: mark a completed maintenance job Operational, and raise a reorder level after a stock-out near-miss.
4. 4.4 Delete: find a specific erroneous movement row first with a SELECT, then delete it by its real MovementID — never delete blindly.
5. Capture your query text and result set (screenshot or copy/paste) for each of the four exercises before moving on.

4.1 Create — new facility and stock receipt

```

-- Add a new depot in Kano (regulator expansion)
INSERT INTO dbo.Facilities
(FacilityCode, FacilityName, FacilityType, City, StateRegion)
VALUES
('FAC-KAN-01', 'Kano Products Depot', 'Depot', 'Kano', 'Kano State');

-- Receive stock: 20 fire-retardant coveralls delivered to Kano Depot
INSERT INTO dbo.StockMovements
(StockItemID, FacilityID, MovementType, Quantity,
ReferenceNote, RecordedBy)
SELECT si.StockItemID, f.FacilityID, 'RECEIPT', 20,
'P0-2026-020 SafetyFirst PPE – Kano opening stock', 'your.initials'
FROM dbo.StockItems si, dbo.Facilities f
WHERE si.SKU = 'SKU-PPE-001' AND f.FacilityCode = 'FAC-KAN-01';

```

4.2 Read — three representative queries

```
-- Q1: HIGH-category PPE items with preferred supplier
SELECT si.SKU, si.ItemName, sup.SupplierName
FROM dbo.StockItems si
JOIN dbo.Suppliers sup ON sup.SupplierID = si.PreferredSupplierID
WHERE si.Category = 'PPE';

-- Q2: stock on hand per item at Warri Products Depot
SELECT si.SKU, si.ItemName, SUM(sm.Quantity) AS QtyOnHand
FROM dbo.StockMovements sm
JOIN dbo.StockItems si ON si.StockItemID = sm.StockItemID
JOIN dbo.Facilities f ON f.FacilityID = sm.FacilityID
WHERE f.FacilityCode = 'FAC-WAR-01'
GROUP BY si.SKU, si.ItemName
ORDER BY si.SKU;

-- Q3: equipment currently Faulty or UnderMaintenance
SELECT e.AssetTag, e.EquipmentName, e.Status, f.FacilityName, f.City
FROM dbo.Equipment e
JOIN dbo.Facilities f ON f.FacilityID = e.FacilityID
WHERE e.Status IN ('Faulty', 'UnderMaintenance');
```

4.3 Update — mark maintenance complete, raise reorder level

```
-- PH Export Terminal Isolation Valve maintenance is complete
UPDATE dbo.Equipment
SET Status = 'Operational', LastInspectionDate = '2026-07-23'
WHERE AssetTag = 'EQ-PHC-0001';

-- Raise the reorder level for gas detectors after a near-miss
UPDATE dbo.StockItems
SET ReorderLevel = 20
WHERE SKU = 'SKU-PPE-003';
```

4.4 Delete — find first, then delete a specific row

```
-- Find the erroneous movement first
SELECT * FROM dbo.StockMovements
WHERE ReferenceNote LIKE '%variance correction%';

-- Then delete the specific row by its real MovementID, e.g.:
-- DELETE FROM dbo.StockMovements WHERE MovementID = 29;
```

IMPORTANT In a regulated environment, deleting transactional history is rarely appropriate — auditors expect an unbroken trail. In real operations, prefer recording an ADJUSTMENT movement (already modeled in the schema) instead of deleting historical rows. The DELETE exercise above is for lab practice only.

Part 5 — Normalization mini-exercise

1. Consider the denormalized flat table a junior developer proposed instead:
StockMovementFlat(MovementID, ItemSKU, ItemName, ItemCategory, SupplierName, SupplierPhone, FacilityName, FacilityCity, MovementType, Quantity, MovementDate).
2. Identify at least three normalization problems with this design — for example, update anomalies if a supplier's phone number changes, repeating ItemName/ItemCategory on every movement row, or an insertion anomaly where you cannot add a new StockItem until its first movement occurs.
3. Map each column in StockMovementFlat to the correctly normalized table it belongs in — StockItems, Suppliers, Facilities, or StockMovements — using the schema you just deployed.
4. Write one sentence explaining why StockMovements is the correct "fact"/transaction table and the other three are "dimension"/reference tables.

Expected result / checkpoints

PetroleumInventoryDB shows the free-offer usage badge in the portal Overview blade.

Azure Data Studio connects successfully and lists all 5 tables under PetroleumInventoryDB > Tables.

Sanity-check counts match 10/8/14/12/30 (growing after your INSERTs).

All four CRUD exercises (4.1–4.4) run without error and return expected rows.

Normalization mini-exercise answers written (3 problems, column mapping, 1-sentence justification).

You disconnected Azure Data Studio/SSMS at the end of the session (cost-safety step).

TIP The first query after any break may be slow — this is the serverless database auto-resuming from an auto-paused state. Wait 10–30 seconds and retry; it is expected behavior, not a fault.

SECTION 06

Implementing Day 3 — DBA & Performance

Day 3 moves learners from "can write SQL" to "can diagnose and fix a production-grade performance problem." The scenario: a Director of Compliance report — "show all HIGH-risk findings at the Warri Products Depot in the last 90 days" — has become slow as the inspection-events table grew to tens of thousands of rows. Learners generate a large synthetic table, capture the execution plan, spot the table scan, fix it with a targeted covering index, and confirm the improvement. The day closes with an observation-only walkthrough of backup/PITR and geo-replication.

What you'll need

1. Completion of the Day 2 lab — an existing PetroleumInventoryDB Azure SQL Database, connected via Azure Data Studio/SSMS.
2. The file labs/datasets/day3_performance_lab.sql from this kit.
3. Basic familiarity with SELECT/WHERE/JOIN from Day 2.

Part 1 — Generate the large table and reproduce the slow query

1. Connect to PetroleumInventoryDB in Azure Data Studio.
2. Open labs/datasets/day3_performance_lab.sql and read the header comment block describing the scenario.
3. Run Step 1 of the script only — it creates the InspectionEvents table and inserts about 12,000 rows using a set-based tally-CTE technique (no cursors or loops needed), plus 4 guaranteed demo rows.
4. Confirm the row count.

Row-count check after Step 1

```
SELECT COUNT(*) AS TotalInspectionRows FROM dbo.InspectionEvents;
-- Expected: 12,004
```

Part 2 — Before: capture the execution plan and identify the scan

1. Before running the next query, enable the execution plan view: click Explain in the query toolbar in Azure Data Studio (in SSMS: Ctrl+M to include the actual execution plan).
2. Run Step 2 of the script — SET STATISTICS IO ON; SET STATISTICS TIME ON; followed by the report query below.
3. Open the execution plan tab and hover over the leftmost/largest operator icon.
4. Confirm you see a Clustered Index Scan (not Seek) with Estimated/Actual Number of Rows Read of about 12,004 — the entire table, even though only a handful of rows actually match.
5. Check the Messages tab for the STATISTICS IO output and note the logical reads count.
6. Record: Before — Operator = Clustered Index Scan; note your logical reads; rows read ≈ 12,004.

The slow report query (Step 2)

```
SELECT
  InspectionID, FacilityCode, FacilityCity, InspectionType,
  InspectorName, InspectionDate, RiskRating, FindingsSummary
FROM dbo.InspectionEvents
WHERE FacilityCode = 'FAC-WAR-01'
AND RiskRating = 'HIGH'
AND InspectionDate >= DATEADD(DAY, -90, '2026-07-23')
ORDER BY InspectionDate DESC;
```

Why this happens: the table's only index is the clustered primary key on InspectionID. There is no index letting SQL Server jump straight to "Warri + HIGH + last 90 days" rows, so the engine reads every row and evaluates the WHERE clause against each one — a full scan.

Part 3 — Diagnose with the missing-index DMV

1. Run the missing-index query from Step 3 of the script (below).
2. Compare the suggestion against the equality/range column design: equality columns (FacilityCode, RiskRating) first, range column (InspectionDate) last, with INCLUDE columns for anything only needed in the SELECT list.
3. Discuss why you should design the index deliberately rather than blindly running the auto-suggested DDL in production — auto-suggestions ignore write-heavy tables, existing similar indexes, and index-maintenance overhead.

Missing-index DMV query (Step 3)

```

SELECT
mid.statement AS TableName,
migs.avg_total_user_cost * migs.avg_user_impact AS EstimatedBenefit,
'CREATE INDEX IX_Suggested ON ' + mid.statement + ' (...)' AS SuggestedIndexDDL
FROM sys.dm_db_missing_index_details mid
JOIN sys.dm_db_missing_index_groups mig
ON mig.index_handle = mid.index_handle
JOIN sys.dm_db_missing_index_group_stats migs
ON migs.group_handle = mig.index_group_handle
ORDER BY EstimatedBenefit DESC;

```

Part 4 — Fix: create the covering index and re-run

1. Run Step 4 of the script to create the covering non-clustered index (below).
2. Re-run the exact same report query from Part 2 (Step 5 of the script) — same column list, same WHERE clause.
3. Open the execution plan again and confirm the operator is now Index Seek, with no Key Lookup operator — meaning the index is "covering" and satisfied the whole SELECT list without a second trip to the clustered index.
4. Check STATISTICS IO again — logical reads should now be a small single- or double-digit number instead of the hundreds seen before.
5. Record: After — Operator = Index Seek; note your new logical reads; rows read = only the matching rows.

The fix — covering non-clustered index (Step 4)

```

CREATE NONCLUSTERED INDEX IX_InspectionEvents_Facility_Risk_Date
ON dbo.InspectionEvents (FacilityCode, RiskRating, InspectionDate)
INCLUDE (FacilityCity, InspectionType, InspectorName,
FindingsSummary);

```

Before/after comparison — fill in your actual numbers

Metric	Before (no index)	After (covering index)
Plan operator	Clustered Index Scan	Index Seek
Rows read	~12,004	a handful (matching rows only)
Logical reads (STATISTICS IO)	hundreds (record your number)	single/double digits (record your number)
Key Lookup present?	N/A	No

Part 5 — Backup, PITR, and geo-replication walkthrough (observe only)

1. In the Azure portal, open your PetroleumInventoryDB resource and select Manage backups. Review the retention policy shown (typically 7 days on the free/serverless tier) and the list of available restore points.
2. Select Restore to open the Point-in-Time Restore wizard, review the Basics tab, then cancel out without confirming — this step is observation only.

3. Go to the database Overview blade and locate Failover groups / Geo-replication (usually under Data management, or via the server-level blade). Open the geo-replication region map.
4. Discuss as a class: for the crude-oil production reporting portal, which secondary region would you nominate for active geo-replication or a failover group, and why — tie this back to the Day 1 region decision matrix.
5. Note that active geo-replication and failover groups typically require a higher, non-free service tier — confirm this in the portal and document it as a future production consideration rather than testing it against the shared lab database.

IMPORTANT Do not actually trigger a geo-restore or failover against the shared lab database — this can disrupt other learners in the same course. This part of the lab is "look, don't click Restore."

Cleanup — required before you finish

1. Drop the InspectionEvents table and its index at the end of the lab (Step 6 of the script) so this temporary practice data does not consume your free-tier 32 GB data allowance long-term.
2. Disconnect Azure Data Studio/SSMS Object Explorer so the database can auto-pause.

Expected result / checkpoints

InspectionEvents table created with 12,004 rows.

"Before" execution plan captured showing Clustered Index Scan with ~12,004 rows read.

Missing-index DMV query run and reviewed.

Covering non-clustered index created successfully.

"After" execution plan captured showing Index Seek with no Key Lookup and far fewer logical reads.

Before/after comparison table completed with your actual session numbers.

Backup/PITR wizard and geo-replication blade both viewed (no restore/failover actually triggered).

InspectionEvents table and its index dropped at the end of the lab.

SECTION 07

Implementing Day 4 — Security Hardening

Day 4 hardens the Day 2 PetroleumInventoryDB database and its application context end-to-end: identity (Microsoft Entra ID, MFA, least-privilege RBAC), encryption at rest and in transit, network access controls, Microsoft Defender for SQL, and auditing. The day closes with the same 24-control security checklist that gets reused, extended, as one of the three Day 5 capstone deliverables — so complete it carefully.

What you'll need

1. Completion of the Day 2 lab — an existing PetroleumInventoryDB on the free-offer serverless tier.
2. Owner or Contributor access on your rg-<initials>-course resource group.
3. The template file labs/datasets/day4_security_checklist.md from this kit.

4. A Microsoft Entra ID user account to designate as the SQL Entra admin (your own course account is fine).

Part 1 — Microsoft Entra ID, MFA, and least-privilege RBAC

1. In the Azure portal, search "Microsoft Entra ID" and open the Microsoft Entra admin center.
2. Go to Identity > Protection > Conditional Access. Review (without necessarily enforcing, if on a shared course tenant) a policy requiring MFA for all users.
3. Note for your checklist: Microsoft requires MFA for all Azure portal access — including Global Admins — starting October 1, 2026. Confirm your own sign-in already uses MFA.
4. Go to your rg-<initials>-course resource group > Access control (IAM) > Role assignments.
5. Select Add > Add role assignment. Choose a scoped role such as SQL DB Contributor (not Owner or Contributor) and assign it to a test/second account if available, to practice least-privilege assignment.
6. Record in your checklist which roles are assigned at the resource-group scope, and confirm no unnecessary Owner assignments exist beyond the course admin.

Source: MFA required for all Azure portal access from October 1, 2026 — <https://techcommunity.microsoft.com/blog/microsoft-entra-blog/microsoft-entra-id-security-updates-what-organizations-need-to-do-now/4522024>

Source: Conditional Access documentation — <https://learn.microsoft.com/en-us/entra/identity/conditional-access/>

Part 2 — Microsoft Entra authentication and database-level least privilege

1. Open your SQL logical server resource (sql-<initials>-course) in the portal.
2. In the left menu, select Microsoft Entra ID (under Settings).
3. Select Set admin, choose your Entra user or group, and Save. This allows Entra-based sign-in going forward, alongside or instead of SQL authentication.
4. Connect to PetroleumInventoryDB in Azure Data Studio using Microsoft Entra authentication this time (change the Authentication type in the connection dialog).
5. In a query window, create a least-privilege application login instead of using the admin account for app connections (see code below).
6. Verify least privilege: confirm the new account is not a member of db_owner.

Create a least-privilege application user

```
-- Run in the context of PetroleumInventoryDB
CREATE USER [app_reporting_user]
WITH PASSWORD = 'UseAStrongUniquePasswordHere!1';
ALTER ROLE db_datareader ADD MEMBER [app_reporting_user];
-- Grant write access only if the app genuinely needs it:
-- ALTER ROLE db_datawriter ADD MEMBER [app_reporting_user];
```

Verify least privilege

```
SELECT dp.name AS RoleName, mp.name AS MemberName
FROM sys.database_role_members rm
JOIN sys.database_principals dp
ON dp.principal_id = rm.role_principal_id
JOIN sys.database_principals mp
ON mp.principal_id = rm.member_principal_id
WHERE mp.name = 'app_reporting_user';
```

Part 3 — Encryption at rest and in transit

1. On the PetroleumInventoryDB resource, go to Transparent data encryption (under Security).
2. Confirm Data encryption is On — TDE is enabled by default on Azure SQL Database, so this is a verification step, not a configuration step.
3. Go to the SQL server resource (not the database) > Networking. Under Encryption in transit, confirm/set Minimum TLS version to TLS 1.2.
4. Run a quick verification query to confirm your current session is encrypted (below).

Verify the current session is encrypted

```
SELECT session_id, encrypt_option
FROM sys.dm_exec_connections
WHERE session_id = @@SPID;
-- encrypt_option should show "TRUE"
```

Part 4 — Network hardening: firewall and Private Endpoint

1. On the SQL server resource, go to Networking.
2. Under Firewall rules, review your existing rule from Day 2 and remove any overly broad rule (e.g. a full 0.0.0.0–255.255.255.255 range).
3. Set "Allow Azure services and resources to access this server" to No, unless a specific in-Azure app tier genuinely needs it — document your decision either way.
4. Under Private access, review the Private endpoint option. Select Add private endpoint to see the configuration wizard, then cancel before completing if you are not proceeding with a VNet in this lab — document it as "reviewed, not implemented due to lab time/cost constraints" if so.
5. If a lab VNet is available, complete the Private Endpoint and set Deny public network access to Yes afterward; otherwise leave public access with the scoped firewall rule and note this clearly in the checklist.

Part 5 — Microsoft Defender for SQL and auditing

1. On the PetroleumInventoryDB resource, go to Microsoft Defender for Cloud (under Security).
2. Select Enable Microsoft Defender for SQL (or confirm it is already enabled at the server level). Use the Free tier or trial window per the cost-safety guardrails.
3. Go to Auditing (under Security). Toggle Azure SQL Auditing to On.
4. Choose a destination for audit logs — select Log Analytics (or a Storage account) and pick/create a workspace inside your rg-<initials>-course resource group.
5. Save, then go to Diagnostic settings (under Monitoring) and confirm a diagnostic setting exists exporting SQLSecurityAuditEvents to your chosen destination.
6. Generate a test audit event by running a simple query (SELECT TOP 5 * FROM dbo.Facilities;), then after a few minutes check the Log Analytics workspace Logs blade for the corresponding audit record.

Part 6 — Azure Policy and patching posture

1. Go to your rg-<initials>-course resource group > Policies (under Governance).
2. Select Assign policy, browse the built-in definitions, and assign one relevant policy — e.g. search "transparent data encryption" or "SQL server auditing" and scope it to your resource group.
3. Note in your checklist that because PetroleumInventoryDB is a PaaS service, Microsoft manages the underlying OS/engine patching automatically — no Update Manager task is needed for the database itself. If your course also stood up a lab VM, confirm Azure Update Manager is enabled for it instead.

Part 7 — Complete the security checklist

1. Open labs/datasets/day4_security_checklist.md.
2. Go through all 24 controls, marking Applies to / Implemented (Y/N) / Evidence-Notes based on what you configured in Parts 1–6. Reference specific portal blades, query results, or screenshot filenames as evidence.
3. Sign off at the bottom of the checklist and note any items marked "N/A" or "Needs rework" with a one-line reason, e.g. "Private Endpoint skipped — no lab VNet available this session."

The 24-control checklist — categories at a glance

Category	Example controls	Scope
Identity & MFA	Entra ID as identity provider; MFA on all sign-ins; Conditional Access policy configured; least-privilege RBAC at resource-group scope	Subscription / Tenant
Database identity	Entra admin set on the SQL server; SQL authentication restricted where possible	Azure SQL logical server
Database least privilege	App account granted only db_datareader/db_datawriter, never db_owner	PetroleumInventoryDB
Encryption at rest	TDE enabled (default, verify status)	PetroleumInventoryDB
Encryption in transit	Minimum TLS 1.2 enforced; "Deny public network access" reviewed	Azure SQL logical server
Network access	Firewall rules scoped to specific IPs; Private Endpoint evaluated; NSG rules reviewed	Server / VNet
Secrets management	Connection strings/passwords stored in Key Vault, access restricted via RBAC	Key Vault
Threat protection	Defender for Cloud at subscription level; Defender for SQL on the server	Subscription / Server
Auditing & logging	Auditing on, logs to Log Analytics/Storage; diagnostic settings exporting SQLSecurityAuditEvents	Server

Category	Example controls	Scope
Governance	Azure Policy assigned to the resource group; resource tagging applied	Resource group
Backup/DR posture	Backups verified, retention noted, PITR test performed; geo-redundancy evaluated	PetroleumInventoryDB
Cost-safety cleanup	Idle connections closed for auto-pause; VMs/App Service plans stopped; resource group scheduled for deletion	Resource group

Expected result / checkpoints

Conditional Access/MFA policy reviewed and confirmed active on your own sign-in.

Resource group IAM shows only least-privilege role assignments (no unnecessary Owner grants).

SQL server has a Microsoft Entra admin configured; you connected successfully via Entra authentication at least once.

app_reporting_user created with only db_datareader (and db_datawriter only if justified) — verified via the role-membership query.

TDE confirmed On; minimum TLS version confirmed at 1.2; encrypt_option = TRUE verified for your session.

Firewall rules scoped to specific IPs only; Private Endpoint reviewed (implemented or explicitly deferred with a documented reason).

Microsoft Defender for SQL enabled; Auditing On with logs flowing; a test audit event located in the logs.

At least one Azure Policy assigned to the resource group.

day4_security_checklist.md fully completed and signed off.

TIP TDE, Always Encrypted, and TLS each protect against a different threat: TDE protects stored data files/backups if physical media is stolen; Always Encrypted keeps specific sensitive columns hidden even from the database engine/DBA view; TLS protects data moving over the network. None of the three substitutes for access control — a legitimate but over-privileged account can still read data the database decrypts for it.

SECTION 08

Implementing Day 5 — The Capstone

Day 5 synthesizes everything: cloud service selection, database administration, and security controls, deployed as one small end-to-end system and defended in a short oral exam. Learners (solo or in pairs) pick one of two regulator scenarios, build a minimal app + managed database + security controls, and submit exactly three deliverables before their oral defense slot.

Business context: you are part of the IT & Digital Transformation unit of a Nigerian upstream/downstream petroleum regulator, under pressure to modernize paper- and spreadsheet-based monitoring while staying compliant with the Nigeria Data Protection Act 2023 (NDPA) and its predecessor guidance, the NDPR. Leadership has mandated that any new system runs on Microsoft Azure, keeps regulated data auditable, encrypted and recoverable, demonstrates data residency awareness, and can be justified to auditors, oversight committees, and the agency's Data Protection Officer.

Step 1 — Choose your scenario (pick exactly one)

1. Scenario A — Downstream Fuel Distribution Monitoring: depot managers and field inspectors log daily petroleum product (PMS, AGO, DPK) receipts, dispatches, and station stock levels so the regulator can monitor supply and flag stock-out/diversion risk in real time.
2. Scenario B — Crude Oil Production & Royalty Reporting: upstream operators submit monthly crude oil production volumes and calculated royalty figures so the regulator can track production against license terms and reconcile royalty obligations.
3. Keep the app deliberately small: a single table/entity, one basic CRUD form, and one summary view is sufficient. This is a security, cloud, and database architecture exercise — not a software engineering exercise.

Azure service mapping for the capstone

Requirement area	Required Azure service	Notes
Compute (app tier)	Azure App Service (PaaS)	Use Free (F1) or Basic (B1) tier for the lab
Managed database	Azure SQL Database	Free-offer serverless General Purpose: up to 10 GP serverless DBs per subscription, 100,000 vCore-seconds + 32 GB data + 32 GB backup free per month
Identity & access	Microsoft Entra ID, RBAC, least privilege	App sign-in via Entra ID; separate roles for submitter vs. reviewer vs. admin
Secrets management	Azure Key Vault	App reads secrets via managed identity, not hardcoded config
Encryption at rest	Transparent Data Encryption (default); Always Encrypted for very sensitive columns	Confirm and document TDE status; do not disable it
Encryption in transit	TLS 1.2+ enforced	Azure SQL requires encrypted connections by default; confirm App Service HTTPS Only
Network protection	NSG and/or Private Endpoint; SQL firewall rules	Minimum: restrict server-level firewall; document what a Private Endpoint would add if deferred
Threat protection	Microsoft Defender for Cloud / Defender for SQL	Enable at subscription or resource level; document findings and remediation status

Requirement area	Required Azure service	Notes
Auditing / logging	Azure SQL Auditing, diagnostic settings, Entra ID sign-in logs	Maps directly to NDPA audit-trail obligations
Patching / configuration	Update Manager for any VM component; App Service auto-managed runtime patching	Document the Microsoft vs. agency patching responsibility split
Backup / recovery	Automated backups + PITR, geo-restore, active geo-replication / failover groups	See the Backup & DR Plan deliverable

Source: Azure SQL Database free offer —

<https://learn.microsoft.com/en-us/azure/azure-sql/database/free-offer>

Step 2 — Build the three required deliverables

1. Deliverable 1: Architecture diagram. Fill in all three parts of the template: (A) a component table listing every Azure service used, its tier/SKU, purpose, and which security boundary it belongs to; (B) the ASCII/text reference architecture showing the request path from a browser through App Service, Key Vault, network controls, and into Azure SQL Database; (C) the Mermaid diagram, rendered in any Mermaid-compatible viewer. All must show the public internet edge, identity boundary, application boundary, secrets boundary, network boundary, data boundary, monitoring/backup boundary, and a data-residency note.
2. Deliverable 2: Completed security checklist. Use the same five domains as Day 4 — Identity & Access, Network Security, Data Protection, Monitoring & Threat Detection, Patching & Secure Configuration. For every control, mark Implemented / Partial / Not Implemented, describe the specific implementation, and reference an evidence file. If a control was not implemented, state the compensating control or justification — do not leave it blank.
3. Deliverable 3: Backup & DR plan. State explicit RPO/RTO targets per component (database, app tier, Key Vault secrets), document the Azure-managed backup schedule and retention window, write out the PITR restore procedure step by step, document your chosen geo-restore/active geo-replication/failover-group approach (or explain why it was deferred), and complete a test-restore record with a real timestamp and result.
4. Perform at least one actual test restore (PITR) where feasible, and record the result, validation method, and time-to-restore compared against your stated RTO — this evidence is worth real points in the rubric.
5. Assign named roles/responsibilities for the backup plan, including a Data Protection Officer / regulatory liaison role, and write a short communication plan for who is notified and who signs off after a restore or failover.

Step 3 — Follow the timeline for the day

1. 09:00–09:30 — Kickoff briefing; choose Scenario A or B; form pairs if applicable.
2. 09:30–11:00 — Provision the Azure SQL Database (free offer) + App Service; build the minimal CRUD/summary app functionality.
3. 11:15–13:00 — Implement security controls: Entra ID sign-in, RBAC, Key Vault, TDE confirmation, TLS/HTTPS Only, firewall/NSG, Defender for Cloud, auditing.
4. 13:45–15:00 — Configure and test backup/PITR; document the geo-restore/failover-group procedure; perform a test restore where feasible.

5. 15:00–16:00 — Assemble the three deliverables using the templates; do an internal peer review by swapping with another pair/individual.
6. 16:15–17:15 — Oral defenses, about 10 minutes per learner/pair.
7. 17:15–17:30 — Wrap-up, course close, resource group teardown reminder.

Step 4 — Submit your work

1. Create a single submission folder named capstone-<your-initials>.
2. Include architecture-diagram.md (or .png/.pdf if hand-drawn/exported), security-checklist.md, and backup-dr-plan.md — all completed from the templates.
3. Include an evidence/ subfolder with screenshots of the App Service, SQL Database, Entra ID role assignments, Key Vault, Defender for Cloud, auditing settings, and a completed backup/restore test.
4. Submit the folder via the agreed course channel (shared drive/LMS) before your scheduled oral defense slot.
5. Be ready to share your screen for the oral defense — you may be asked to show a live Azure portal blade, not just a screenshot.

IMPORTANT Cost ceiling for the capstone: Free/Basic tiers only. Azure SQL Database must use the free-offer serverless configuration; App Service must use F1 or B1. No Business Critical, Premium, or multi-region tiers. Use synthetic/sample data only — never real licensee, depot, or operator data.

— How instructors grade the capstone —

Grading rubric summary — 100 points, 70/100 to pass

Deliverable	Points	Section minimum to pass
1. Architecture Diagram	20	≥60% of 20 (≥12)
2. Security Checklist	30	≥60% of 30 (≥18)
3. Backup & DR Plan	25	≥60% of 25 (≥15)
4. Oral Defense	25	≥60% of 25 (≥15)

Pass rule

- A learner passes only if the total score is at least 70/100 AND every one of the four sections independently meets its 60% minimum.
- Scoring below the minimum in any single section fails the capstone regardless of total score, and that section must be remediated.
- Cross-cutting deductions: up to 3 points overall if the learner selected tiers beyond the Free/Basic ceiling without justification, or showed no cost awareness (e.g. no mention of auto-pause).
- Regulatory alignment: full marks in Data Protection and RPO/RTO justification require explicit NDPR/NDPA, data sovereignty, auditability, or availability framing specific to the oil & gas regulator scenario — generic answers are capped at "Good."
- Evidence over assertion: any control claimed "Implemented" without a corresponding screenshot is scored as "Partial," not "Implemented."

— Running the oral defense —

Step-by-step: how to run each 10-minute defense

1. Have the learner(s) share their screen with the live Azure portal open — not just static screenshots — so you can spot-check at least one control live.
2. Give the learner 60 seconds to summarize: scenario chosen, what was built, one thing they are proud of.
3. Ask 4–6 questions from the question bank, drawing at least one from each of the four categories: Cloud & Architecture, Database, Security, Regulatory & Governance.
4. Ask one unscripted follow-up/curveball based on something specific in their own submission — this is your primary signal for distinguishing "Good" from "Excellent," since core questions can be rehearsed in advance.
5. If a pair is presenting, require both members to answer at least 2 questions each — do not let one person answer everything.
6. Give immediate verbal feedback and record the score against the rubric's Section 4 (Oral Defense).

Sample oral-defense questions by category

Category	Sample question	What a good answer contains
Cloud & Architecture	Why did you choose Azure App Service instead of a virtual machine for the app tier?	PaaS vs. IaaS tradeoff: App Service offloads OS/runtime patching, deploys faster, has built-in scaling/HTTPS — appropriate for a small regulatory app
Database	Explain the difference between your PITR restore and a geo-restore. When would you use each?	PITR restores within the same region to any point in the retention window for logical errors; geo-restore restores from geo-redundant backup to a different region for a regional outage — different RPO/RTO
Security	Where are your database credentials actually stored, and who/what can access them?	Stored in Key Vault, accessed via the App Service's managed identity — not hardcoded, not in source, not in plaintext env vars
Regulatory & Governance	What would you tell the agency's DPO about where this data physically resides, and what would need to happen before this went into production?	Names the region, acknowledges any cross-border residency implication, and identifies realistic next steps — DPIA, DPO/legal sign-off, NDPA compliance review — rather than claiming the POC is already compliant

IMPORTANT Red flag during the defense: a learner who claims the lab build is fully NDPA-compliant with no caveats. The correct, well-scored answer always distinguishes a training proof-of-concept from a production-ready, fully compliant system.

SECTION 09

Instructor Delivery Guide

This section condenses the full Facilitation Plan into a working rhythm you can actually follow live. It assumes you have already completed Section 3's one-time setup and read each day's section above. Use this as your control panel during the week; go back to the full 01_Facilitation_Plan.md for talking points, analogies, and misconception call-outs.

The daily rhythm (every day, 09:00–17:00)

Block	Typical timing	Purpose
Recap + exit-poll review	09:00–09:15/09:20	Address yesterday's misconceptions before moving on
Teaching blocks (2–3 per day)	Morning through early afternoon	Lecture + live demo, regulator analogies throughout
Short breaks	Mid-morning and mid-afternoon	Protect these — attention dips noticeably by mid-morning
Lunch	Midday, ~1 hour	Fixed block; confirm catering timing in advance
Hands-on lab (the core of the day)	Afternoon, 2–3.5 hours	This is where the real learning happens — circulate constantly
Lab debrief	Late afternoon	Compare results across pairs/pods; surface trade-off thinking
Wrap & checkpoint: exit poll + preview next day	16:50–17:00	3 quick questions, show-of-hands or sticky notes — instructor signal, not grading

Per-day instructor prep, condensed

Day	Night-before prep priority
Day 1	Pre-provision your own demo subscription with one stopped VM and one paused Azure SQL DB to show IaaS vs. PaaS live. Print the case-study handout. Know the current African/EMEA region facts cold.
Day 2	Confirm every learner's Azure SQL Database is created before the 11:45 connection slot. Pre-test the exact Azure Data Studio/SSMS connection steps on the actual training-room network. Have the local Docker fallback ready.
Day 3	Rehearse the before/after execution-plan demo end-to-end — timing and exact query text matter for a clean contrast. Review current backup/geo-replication portal navigation for interface drift.

Day	Night-before prep priority
Day 4	Pre-verify TDE is on by default in your demo database. Pre-create an Entra test user so the RBAC demo does not require live signup. Re-confirm the current MFA enforcement date and Conditional Access navigation path.
Day 5	Pre-build and time the full reference architecture yourself within 48 hours. Confirm the oral-defense rubric was communicated at 09:20, not for the first time at 16:00. Pre-plan weaker/stronger pairings before the day starts.

Engagement tactics that work across the week

- Day 1: pair a manager (business framing) with a sysadmin/DBA (technical depth) for the case study — this mirrors how real recommendations get made.
- Day 2: pair a stronger SQL learner with a weaker one using a 'navigator explains, driver types' role split, and celebrate the first successful `SELECT * FROM` result out loud.
- Day 3: pair a DBA-leaning learner with a sysadmin/manager, and ask the DBA to narrate their reasoning out loud; use a friendly 'before vs. after query time' competition at debrief.
- Day 4: rotate which persona's framing you lead with (manager/compliance, DBA/encryption, sysadmin/network) so no one checks out for a full block; use a 2-minute 'how would you attack this?' mini-activity before the hardening lab.
- Day 5: pair each weaker learner with a stronger one but make the weaker learner the lead presenter for the oral defense — this forces real understanding transfer, not silent carrying.

Contingency: connectivity or Docker fallback pivot script

1. If Azure resource creation is slow, keep teaching the concept while resources provision in the background — never let the room sit idle watching a spinner. If delays exceed about 10 minutes for multiple learners, switch that pod to the local Docker fallback for the remainder of the lab.
2. If the free-tier compute/storage limit is hit mid-course, explain the auto-pause behavior and set the database to "Auto-pause the database until next month" rather than upgrading to a paid tier.
3. If internet connectivity drops (Lagos power/fibre issues), pivot the whole room immediately: "We'll continue locally so no one loses lab time; everything about the exercise is identical, only the connection target changes." Switch connection strings from `<server>.database.windows.net` to `localhost` per the local lab guide variant.
4. If a corporate firewall blocks port 1433/5432, use the local Docker fallback for that learner/pod immediately, then escalate to the network checklist owner so it is fixed before the next day.
5. If a learner's laptop cannot run Docker, pair them with a neighbor who can, using a shared-screen driver/navigator setup, or use your own instructor machine as a shared demo target for that pair only.

TIP Keep brief 2–3 sentence notes per learner after each exit poll and lab circulation. These notes drive your Day 5 pairing decisions and let you flag anyone falling behind early enough to intervene — an extra 15 minutes at lunch, a targeted example — rather than discovering a gap for the first time during the capstone.

IMPORTANT Do not record open Q&A sessions if learners discuss internal agency systems — regulator content can be sensitive. Keep specific internal-system discussion for offline conversations, not recorded open-floor Q&A, especially on Day 4.

SECTION 10

Troubleshooting & FAQ

This section compiles the most common issues encountered across every lab in this kit into one reference. If something breaks during a session, check here first before escalating.

Common issues and fixes, by day

Symptom	Likely cause	Fix
"Cannot connect: Client with IP address 'x.x.x.x' is not allowed to access the server"	Your current client IP is not on the SQL server's firewall allow-list	Go to the SQL server's Networking blade in the portal, add your current client IP, then retry the connection
First query after a break is slow or times out	The serverless database is auto-resuming from an auto-paused state	Wait 10–30 seconds and retry — this is expected behavior, not a fault
INSERT fails with a FOREIGN KEY constraint conflict	You referenced a FacilityID/StockItemID/SupplierID that does not exist yet	Run a SELECT on the parent table first to get a valid ID
Free-offer usage exceeded for the month	You hit the 100,000 vCore-seconds or 32 GB data/backup monthly limit	Confirm "Auto-pause the database until next month" in the portal prompt rather than upgrading to a paid tier
Execution plan still shows a scan after creating the index (Day 3)	You re-ran a different query than the one captured in Part 2	Confirm you ran the exact query text — same column list, same WHERE clause — otherwise a Key Lookup or scan can reappear
STATISTICS IO output not visible (Day 3)	Looking in the wrong pane	In Azure Data Studio, check the Messages tab next to Results; in SSMS it's the Messages sub-tab of the results pane
"CREATE INDEX" fails with a duplicate name error (Day 3)	You already created the index in a previous run	Run DROP INDEX IX_InspectionEvents_Facility_Risk_Date ON dbo.InspectionEvents; first, or skip ahead to Part 4 Step 2
Portal "Restore" or "Failover groups" options greyed out (Day 3)	Expected on the free-offer serverless tier for some geo-replication features	Not a blocker — proceed with the class discussion instead of trying to enable it

Symptom	Likely cause	Fix
"Set admin" for Microsoft Entra ID greyed out on the server (Day 4)	You lack Global Administrator or the Microsoft.Sql/servers/administrators/write permission	Ask your instructor for elevated access if using an organizational tenant
Entra authentication connection fails in Azure Data Studio (Day 4)	The Entra account signing in has not been granted access inside the database	Run <code>CREATE USER [user@domain.com] FROM EXTERNAL PROVIDER</code> ; or use the designated server admin account
Auditing logs not appearing in Log Analytics after several minutes (Day 4)	Wrong diagnostic category enabled, or no qualifying activity generated yet	Confirm the <code>SQLSecurityAuditEvents</code> category (not just <code>SQLInsights</code>) is checked, and generate a <code>SELECT/DML</code> after enabling it
Defender for SQL shows "Not enabled" right after enabling it (Day 4)	Propagation delay, or enabled at the wrong scope	Wait a few minutes and refresh; confirm it was enabled at the server level, not just the database level
Private Endpoint wizard requires a VNet you don't have (Day 4)	No lab VNet provisioned	Document it as deferred in the checklist rather than building a full VNet under time pressure
Learners can't create resources — permissions/quota error (any day)	Subscription not fully activated, region restricted for free tier, or free-offer database count (10 GP serverless) already reached	Check the exact error message; try a different supported region, or pair with a learner whose environment works and follow up individually at the break
Room AV/projector failure	Hardware issue	Continue via learners' own screens with you narrating and sharing exact commands in a shared doc/chat; use the whiteboard for diagrams

Frequently asked questions

- Q: Do I need to know SQL before this course? A: It helps but is not mandatory. Day 2 is built assuming some learners have never written SQL, and the facilitation plan includes specific pairing tactics for exactly this.
- Q: What happens if my free-tier quota runs out mid-lab? A: You will see an auto-pause prompt — accept it. Do not upgrade to a paid tier; work with a partner or the local Docker fallback for the rest of that session.
- Q: Is there really no Azure region in Nigeria? A: Correct, as of this course. This is a deliberate teaching point in the Day 1 lab — the correct response is a documented compliance justification, not silently picking any region.
- Q: Can I use PostgreSQL instead of Azure SQL Database? A: Yes — every lab in this kit has a PostgreSQL alternative note with the equivalent free-tier facts and syntax differences. Azure SQL Database remains the primary path.

- Q: What if my capstone pair disagrees on the architecture? A: Document both perspectives in the deliverable if needed, but you must be able to defend the choices made during the oral defense regardless of who typed what.
- Q: Does TDE alone make our data NDPA-compliant? A: No. TDE is one control among many. Compliance requires the full set — encryption, access control, auditing, data minimization, and a documented data-residency rationale — not any single technical control in isolation.

SECTION 11

Appendix: Cost Management, Cleanup & Glossary

Two last things every learner and instructor must do: keep costs at zero throughout the week, and leave nothing running afterward. This appendix also defines the key technical terms used throughout this handbook, for readers newer to cloud, database, or security concepts.

Daily cost-safety routine (repeat every day)

1. Disconnect Azure Data Studio/SSMS Object Explorer whenever you step away from your machine, so the serverless database can auto-pause.
2. Stop or deallocate any VMs and App Service plans you created for testing, at the end of every day — do not leave them running overnight.
3. If you see a free-tier limit warning, choose "Auto-pause the database until next month" — never upgrade to a paid tier to work around a lab quota limit.
4. Confirm your resources are still tagged course=cdds and living only inside your rg-<initials>-course resource group — never spread lab resources across multiple resource groups.

End-of-course cleanup (do this on Day 5, before you leave)

1. Stop/deallocate any App Service plans and VMs one final time.
2. In the Azure portal, go to Resource groups and open rg-<initials>-course.
3. Confirm the resource list matches what you expect — no orphaned resources from earlier in the week.
4. Select Delete resource group, type the resource group name to confirm, and select Delete.
5. Wait for the deletion to complete, then verify the resource group no longer appears in your Resource groups list.
6. If you used a shared/organizational subscription, confirm with your instructor or IT admin that the deletion has fully propagated before considering the course closed out.

IMPORTANT Deleting the resource group is permanent and removes every resource inside it, including the PetroleumInventoryDB database and all lab data. Do not delete it before your Day 5 oral defense is complete and your capstone submission has been made.

Glossary of key terms

Term	Plain-language definition
IaaS (Infrastructure as a Service)	You rent raw compute/storage/network (e.g. a VM) and manage the OS, patching, and scaling yourself.
PaaS (Platform as a Service)	The provider manages the OS, runtime, and infrastructure; you manage only your application and data (e.g. Azure SQL Database, Azure App Service).
RDBMS	Relational Database Management System — software that stores data in structured tables with defined relationships (e.g. Azure SQL Database, PostgreSQL).
Normalization	Organizing data into related tables to remove duplication and update risk, at the cost of needing JOINS to bring related data back together. Practical schemas often stop at 3NF (Third Normal Form).
Index	A separate, ordered structure that lets the database engine find rows without scanning the whole table — like the index at the back of a book.
TDE (Transparent Data Encryption)	Encrypts the database's data files and backups at rest so stolen physical media cannot be read; enabled by default on Azure SQL Database.
RBAC (Role-Based Access Control)	Assigning permissions through defined roles (e.g. SQL DB Contributor) scoped to the smallest resource needed, rather than broad Owner/Contributor access.
MFA (Multi-Factor Authentication)	Requiring a second proof of identity beyond a password. Microsoft requires MFA for all Azure portal access, including Global Admins, from October 1, 2026.
Conditional Access	A Microsoft Entra ID feature that applies access rules (such as requiring MFA) based on risk signals like location, device compliance, or sign-in risk, configured in the Microsoft Entra admin center under Identity > Protection > Conditional Access.
PITR (Point-in-Time Restore)	Restoring a database to any point within its retention window, always into a new database, to recover from logical errors like a bad update or accidental delete.
RPO (Recovery Point Objective)	How much data you can afford to lose, measured in time — addressed primarily by backup frequency and PITR granularity.
RTO (Recovery Time Objective)	How quickly you must be back online after an incident — addressed primarily by geo-replication and failover groups, not backups alone.
NDPR / NDPA	Nigeria Data Protection Regulation and its successor, the Nigeria Data Protection Act 2023 — the legal framework governing personal and regulated data handling in Nigeria, central to this course's compliance framing.

Source: Azure SQL Database free offer —

<https://learn.microsoft.com/en-us/azure/azure-sql/database/free-offer>

Source: Azure Database for PostgreSQL Flexible Server free account deployment — <https://learn.microsoft.com/en-us/azure/postgresql/flexible-server/how-to-deploy-on-azure-free-account>

Source: Microsoft Entra ID security updates — MFA required for all Azure portal access from October 1, 2026 — <https://techcommunity.microsoft.com/blog/microsoft-entra-blog/microsoft-entra-id-security-updates-what-organizations-need-to-do-now/4522024>

Source: Microsoft Entra Conditional Access documentation —

<https://learn.microsoft.com/en-us/entra/identity/conditional-access/>

TIP Keep this handbook alongside the repository's own README_START_HERE.md and 01_Facilitation_Plan.md for the week — this handbook summarizes and sequences them, but the source files remain the definitive reference for exact wording, screenshots, and edge cases.